

LIHTC KG-RAG

Knowledge Graph RAG for Tax Credit Mortgage Underwriting

Technical Architecture & Team Briefing Document

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Prepared for engineering, data, and domain expert team review

1. Purpose & Scope

This document captures the complete architecture, design decisions, and domain context for a Knowledge Graph RAG (KG-RAG) system applied to USA Tax Credit Program mortgage underwriting in the real estate domain. It is intended as the primary onboarding reference for the engineering and data science team before detailed sprint planning begins.

Objective

Build a KG-RAG capability that can assist underwriters in evaluating LIHTC (Low Income Housing Tax Credit) mortgage deals by:

- Populating structured underwriting memo sections from public data sources
- Reasoning across a regulatory knowledge graph using multi-hop traversal
- Generating cited, auditable rationale grounded in §42 regulatory hierarchy
- Benchmarking deal-level analysis against peer community context

1.1 What this is NOT

- Not a product — this is a capability / proof-of-concept for internal underwriting use
- Not a replacement for human underwriters — it assists and accelerates the workflow
- Not a compliance system — outputs must be reviewed by qualified professionals
- Not dependent on any existing commercial platform — built independently on public data

1.2 Domain

LIHTC (the Low Income Housing Tax Credit) is the primary federal mechanism for financing affordable rental housing in the USA, governed by IRC §42. Mortgage underwriting in this domain is complex because it involves a layered regulatory hierarchy, income-based rent restrictions, compliance obligations over 15–30 years, and a tax credit equity market. The combination of structured public data availability and domain complexity makes it an ideal KG-RAG application.

2. Domain Knowledge — LIHTC Regulatory Hierarchy

The single most important domain concept for the team to internalize is that LIHTC is governed by a strict precedence hierarchy. This hierarchy is not just background knowledge — it is encoded directly into the knowledge graph as directed edges, and the RAG retrieval follows this hierarchy in order. A state rule cannot override a federal statute; a GSE guide cannot override an IRS notice.

Level	Authority	Key documents & rules
1 (highest)	Federal statute	IRC §42, §142, §145 — the primary LIHTC statute; cannot be overridden by anything below
2	IRS administrative guidance	Revenue Procedures, PLRs, TAMs, IRS Notices, Revenue Rulings — interpret and implement §42
3	GSE program rules	Fannie Mae DUS guide, Freddie Mac TAH program, FHA 223(f) — set underwriting parameters for insured/guaranteed loans
4	State QAP	Qualified Allocation Plans — each state HFA publishes annually; governs credit allocation scoring, set-asides, income targeting
5 (deal level)	Deal documents	LURA (Land Use Restriction Agreement), Form 8609, Form 8586, Loan Agreement — property-specific binding obligations

2.1 Key LIHTC concepts the team must understand

- QCT (Qualified Census Tract): HUD-designated high-poverty tracts that grant a 30% basis boost to eligible basis — a major financial benefit that the graph must detect automatically
- DDA (Difficult Development Area): Similar to QCT; HUD-designated areas with high construction costs — also grants the 30% basis boost
- AMI (Area Median Income): HUD publishes annually by MSA and county; determines maximum allowable rents and tenant income limits. The year matters — using 2022 limits on a 2023 deal is a material error
- DSCR (Debt Service Coverage Ratio): Net Operating Income / Annual Debt Service. GSE minimum is typically 1.20; FHA is 1.176. The eval framework must check this
- Credit period: 15 years for federal compliance; many deals have extended use agreements (30 years) via state LURAs
- Form 8823: IRS noncompliance notification filed by state HFAs — a compliance event that the graph tracks as a node
- 9% vs 4% credit: 9% credits are competitively allocated via QAP scoring; 4% credits are non-competitive but require private activity bond financing. Different underwriting risk profiles

3. Data Strategy — Public vs Simulated

A foundational design decision: we do not need proprietary loan files to build and validate this capability. The majority of the knowledge graph can be seeded from entirely public data. Only deal-level financial specifics require simulation, and those simulations are strongly constrained by §42 structural rules.

3.1 Publicly available data (free, structured, downloadable)

Source	Data available	Use in KG
HUD LIHTC Database	Every LIHTC project placed in service 1987–present: address, units, AMI targeting, credit type, allocation year, developer name, placed-in-service date, census tract	Primary Property and TaxCreditProgram node source — thousands of real deals
HUD AMI Tables	Annual income limits by MSA/county, household size, AMI percentage (50%, 60%, 80%)	AMILimit nodes — critical for rent calculation and set-aside verification
HUD QCT/DDA Lists	Annual HUD designation of Qualified Census Tracts and Difficult Development Areas by census tract FIPS code	QCT and DDA flags on CensusTract nodes — drives basis boost rule
HUD Fair Market Rents	Annual FMR by MSA and bedroom size	Market rent baseline for Section 5 of UW memo
IRS.gov	Full text of IRC §42, Revenue Procedures (esp. 2003-82), IRS Notices, Form instructions for 8609/8586/8823	Regulation nodes — the statutory backbone of the graph
State HFA websites	Qualified Allocation Plans (QAPs) published annually — 50 states, each different	QAPRule nodes — state-specific scoring, set-asides, income targeting requirements
FFIEC / HMDA	Loan-level origination data by census tract; CRA exam reports	Market context and lender behavior in target geographies
Fannie/Freddie guides	DUS Seller/Servicer Guide, TAH program documentation — public PDFs	GSE rule nodes: DSCR floors, LTV caps, reserve requirements
FHFA / Census ACS	House price index, conforming limits, tract demographics	CensusTract enrichment nodes

3.2 Data that must be simulated

These items do not exist publicly at the loan level. However, they are not free-form fiction — each is structurally constrained by §42 rules, giving us guardrails for realistic synthesis.

Simulated item	Constraint / guardrail	Why it can't be public
Loan origination file	LTV capped by GSE program (e.g., 87% FHA); DSCR must exceed 1.20; loan amount derived from NOI	Borrower-level financial data is private
Sources and uses / capital stack	Equity = credit allocation × \$/credit price; debt sized to DSCR floor; dev fee capped by state QAP (typically 15%)	Deal-level financial structure is proprietary
15-year financial pro forma	Rent escalation capped at HUD AMI changes; vacancy typically 5–7% for affordable; OpEx ratios standard by property type	Lender internal models are proprietary
Tax credit equity pricing	\$/credit \$ moves with rate environment; typically \$0.85–\$0.95 in 2023; IRR targets 5–7% for institutional investors	Investor pricing is negotiated and private
Compliance event history	8823 noncompliance events: frequency, type (income, rent, habitability), cure timeline — simulated per §42 standards	IRS/HFA compliance records are not public at deal level
Sponsor financial profile	Net worth, liquidity, prior deal count — can be inferred directionally from HUD DB developer name lookup	Borrower financials are private
Market rent comparables	HUD FMR provides floor; simulate spread above FMR based on MSA market conditions	Granular comp data is proprietary (CoStar, etc.)
Stress scenarios	Rate shocks, rent freeze, vacancy spikes, recapture events — parameterized against §42 recapture rules	Scenario modeling is internal to lenders

Design decision: anchor simulations to real HUD properties

The recommended approach is to pick real properties from the HUD LIHTC database (identified by `hud_id`) and use their actual public attributes (location, tract, credit type, AMI targeting, placed-in-service date) as the foundation.

Then layer synthetic financial data on top. This gives you realistic geography, real QCT/DDA flags, real AMI limits, and real regulatory obligations — while the financials are simulated within known structural bounds.

Result: a training and test corpus that is factually correct on all public-domain facts and structurally realistic on all simulated facts.

4. Underwriting Memo Structure & Fillability

A standard LIHTC mortgage underwriting memo has 13 sections. The KG-RAG system's primary output is structured text for each section, with every factual claim tagged to a source node in the graph. The table below maps each section to its fill strategy.

#	Section	Fill strategy	KG node types involved
1	Property description & location	PUBLIC DATA	Address, units, year built, census tract, QCT/DDA flag — HUD LIHTC DB + FFIEC tract data Property, CensusTract
2	Tax credit program summary	PUBLIC DATA	Credit type (9%/4%), basis, placed-in-service date, credit period — HUD LIHTC DB + IRS §42 TaxCreditProgram, Regulation
3	Income targeting & AMI set-asides	PUBLIC DATA	% units at 50%/60% AMI, income limits by bedroom and household size — HUD AMI tables AMILimit, SetAside
4	Maximum allowable rents	PUBLIC DATA	Computed: HUD AMI × 30% / 12 by bedroom type — fully derivable from public data RentLimit, UnitMix
5	Market rent comparables	PUBLIC + SIMULATE	HUD FMR provides market rent floor; simulate spread above FMR by MSA conditions MarketRent, Comparable
6	Compliance & regulatory history	PUBLIC + SIMULATE	State HFA audit reports (partial); simulate Form 8823 noncompliance event timelines ComplianceEvent, Form8823
7	Sponsor / developer experience	PUBLIC + SIMULATE	HUD DB tracks developer name across deals (prior project count); simulate financials Sponsor, Developer
8	Sources & uses / construction budget	SIMULATE	§42-constrained: equity = credit × pricing; debt at DSCR floor; dev fee capped by QAP CapitalStack, LoanFile
9	Financial projections (DSCR/NOI/LTV)	SIMULATE	15-yr pro forma with standard assumptions: 5% vacancy, CPI rent escalation, OpEx ratios ProForma, LoanTerms
10	Tax credit equity pricing	SIMULATE	Market \$/credit \$ benchmarks; investor IRR targets; syndicator fee structure InvestorPricing, Equity
11	Regulatory compliance checklist	PUBLIC DATA	§42 minimum set-aside tests, state QAP requirements, fair housing, LURA term — fully public Regulation, QAPRule, LURA
12	Risk factors & mitigants	PUBLIC + SIMULATE	Recapture risk (§42 rules), rate risk, vacancy risk — rules public; scenario data simulated RiskFactor, Scenario
13	UW recommendation & conditions	KG-RAG OUTPUT	Generated by LLM from graph context — approve/condition/decline with cited §42 node references Decision, Rationale

Legend:

PUBLIC DATA = fully derivable from public sources. **PUBLIC + SIMULATE** = public skeleton + synthetic fill. **SIMULATE** = synthetic, §42-constrained. **KG-RAG OUTPUT** = generated by the system.

5. Finalized System Architecture

The architecture consists of seven layers. The diagram below describes the complete flow from raw source documents through to underwriting memo output and evaluation feedback.

Layer 0 Domain knowledge + per-node & per-edge embedding ★ <i>NEW — not in original diagram</i>
Layer 1 Ingestion — NER, coreference, entity linking, dual index write
Layer 2 Knowledge graph + community detection (Leiden/Louvain) ★ <i>NEW — not in original diagram</i>
Layer 3 Query planning + decomposition + routing ★ <i>NEW — not in original diagram</i>
Layer 4 Three-stream retrieval — graph + semantic + community ★ <i>NEW — not in original diagram</i>
Layer 4b Context reranker + conflict resolver + deduplication ★ <i>NEW — not in original diagram</i>
Layer 5 LLM generation — UW memo section output with cited node refs
Layer 6 Eval framework — graph retrieval + rule accuracy + conclusion + citation ★ <i>NEW — not in original diagram</i>

Gaps identified vs original diagram — all now addressed
★ Layer 0: Per-node and per-edge embeddings were absent. Without these, semantic search over graph structure is impossible.
★ Layer 2: Community detection (Leiden clustering) was absent. Communities enable global 'peer cohort' context that traversal alone cannot provide.
★ Layer 3: Query decomposition and routing was absent. Complex UW questions must be broken into typed sub-queries before retrieval.
★ Layer 4: Only graph search fed the LLM. Three streams are required: graph traversal + semantic kNN + community summaries.
★ Layer 4b: No context reranker existed. Without it, conflicting facts from different streams reach the LLM unresolved.
★ Layer 6: Eval framework was absent from the architecture entirely.

6. Layer-by-Layer Detail

Layer 0 — Domain Knowledge + Embedding

This layer has two responsibilities. First, it defines the LIHTC domain ontology — the typed vocabulary of entities and relationships that the graph will use. Second, it generates embeddings for every node type and every edge type before ingestion begins. These embeddings are what enable semantic search over graph structure, not just over document text.

Regulatory hierarchy (encoded as directed edges, not just separate nodes)

- Federal statute (IRC §42) → IRS guidance → GSE program rules → State QAP → Deal documents
- Precedence is structural: child nodes inherit constraints from parents; child rules cannot contradict parent rules
- Graph traversal must respect this direction when resolving conflicting rules

Per-node embedding

- Every entity node (Property, TaxCreditProgram, AMILimit, Regulation, Sponsor, etc.) is embedded independently using a domain-fine-tuned encoder
- This allows semantic kNN queries like 'find rules similar to basis boost eligibility' even if the exact label is unknown
- Recommended model: text-embedding-3-large (OpenAI) or a fine-tuned version on §42 text

Per-edge embedding

- Each relationship type (SUBJECT_TO, LOCATED_IN, RECEIVES_CREDIT, HAS_EVENT, etc.) is typed and embedded
- This enables edge-type-aware retrieval — a query about compliance obligations only traverses SUBJECT_TO and HAS_EVENT edges, not LOCATED_IN

Layer 1 — Ingestion

Standard NLP pipeline applied to the source document corpus. The key LIHTC-specific requirements are:

- Entity disambiguation: 'California' as a state vs. 'California HFA' as an organization must be resolved
- Temporal entity linking: AMI limits are year-specific — a 2022 limit and a 2023 limit for the same MSA are different nodes
- Regulatory cross-referencing: IRS notices frequently reference specific §42 subsections — these cross-references become graph edges
- Dual index write: every extracted entity goes to both the graph store (Neo4j or Kuzu) and the vector store (Pinecone or Weaviate) simultaneously

Layer 2 — Knowledge Graph + Community Detection

Graph ontology — node types

Node type	Description and key attributes
Property	Real LIHTC property from HUD DB: hud_id, address, census_tract, placed_in_service_year, total_units, credit_type
TaxCreditProgram	Credit program instance: credit_type (9%/4%), allocation_year, credit_amount, compliance_period
AMILimit	HUD annual limit: year, msa_code, bedroom_count, ami_pct (50/60/80), max_income, max_rent
CensusTract	Tract-level geo node: fips_code, qct_flag, dda_flag, state, county, msa
Regulation	Statutory or guidance text node: citation (e.g., §42(g)(1)), source (IRS/GSE/HFA), text_chunk, effective_date
QAPRule	State QAP rule: state, year, rule_type (scoring/set-aside/targeting), rule_text, points_value
Sponsor	Developer/owner entity: entity_name, prior_deal_count (from HUD DB), state
ComplianceEvent	Form 8823 or audit event: event_type, event_date, cure_date, resolution_status
LoanFile (simulated)	Deal-level loan: loan_amount, dscr, ltv, interest_rate, amortization, origination_date
CapitalStack (simulated)	Sources and uses: equity_amount, debt_amount, grants, deferred_dev_fee, credit_equity_price
RiskFactor	Identified risk: risk_type, severity, mitigant_description

Graph ontology — relationship types

Relationship	Meaning
LOCATED_IN	Property → CensusTract: geographic placement
IS_QCT / IS_DDA	CensusTract → [boolean flag]: basis boost eligibility
RECEIVES_CREDIT	Property → TaxCreditProgram: which program the property participates in
SUBJECT_TO	Property / LoanFile → Regulation: which rules apply
HAS_AMI_LIMIT	TaxCreditProgram → AMILimit: applicable income and rent limits
GOVERNED_BY	TaxCreditProgram → QAPRule: state QAP rules governing allocation
ORIGINATED_BY	LoanFile → Sponsor: who developed/owns the deal
HAS_EVENT	Property → ComplianceEvent: compliance history timeline
PRICED_AT	TaxCreditProgram → CapitalStack: equity pricing for the credit

PRECEDES (hierarchy)	Regulation → Regulation: encodes the regulatory precedence chain
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Community detection

- Algorithm: Leiden clustering (preferred over Louvain for resolution stability)
- Community examples that naturally form: CA 9% deals 2018–2022 under similar QAP rules; NYC 4% bond deals with FHA 223(f) financing; rural USDA 515 + LIHTC hybrid deals
- Pre-computed community summaries: for each community, an LLM generates a ~500-word summary covering typical DSCR ranges, common compliance risks, market rent spreads, and peer sponsor profiles
- These summaries are stored as flat text in a separate community summary index and retrieved by the community retrieval stream

Layer 3 — Query Planning + Decomposition

A real underwriting question is almost never a single-stream query. The query planner uses an LLM to decompose the question before retrieval begins.

Example decomposition:

Input query: "Does this property qualify for the QCT basis boost, and what are the maximum allowable rents for 2BR units at 60% AMI?"

Sub-query	Routed to
Sub-query 1: Is census tract [X] designated as a QCT by HUD?	Graph search: Property → CensusTract → IS_QCT flag → §42(d)(5)(B) rule node
Sub-query 2: What is the 2023 HUD income limit for 2BR at 60% AMI in MSA [Y]?	Semantic search: kNN on AMILimit node embeddings for this MSA, year, bedroom, and AMI %
Sub-query 3: What is the typical DSCR and compliance profile for deals like this?	Community retrieval: match to peer cohort community summary by state, credit type, and vintage

- The planner also classifies each sub-query as local (specific deal fact) or global (peer context) — local queries go to graph + semantic; global queries go to community
- Unanswerable sub-queries (data not in graph) are flagged explicitly rather than hallucinated

Layer 4 — Three-Stream Retrieval

Stream 1: Graph traversal

- Uses Cypher (Neo4j) or equivalent graph query language
- Multi-hop: can traverse Property → CensusTract → QCT → §42(d)(5)(B) in a single query
- Respects the regulatory hierarchy direction — traversal from deal up to statute, not sideways
- Returns: a subgraph — the set of nodes and edges relevant to the sub-query, with node IDs for citation

Stream 2: Semantic / vector search

- kNN query over the per-node embedding index
- Handles vocabulary mismatch: query says 'income restriction' → finds AMILimit node despite different label
- Also retrieves relevant §42 text chunks from the vector store that weren't captured as graph nodes
- Returns: ranked list of nodes + text chunks with similarity scores

Stream 3: Community retrieval

- Matches the deal to its community (by state, credit type, vintage, QAP cohort)
- Retrieves the pre-computed community summary text
- Provides: peer benchmarks (typical DSCR, vacancy, compliance rates), regional risk norms, historical patterns
- Returns: structured community summary (~500 words) tagged with community ID

Layer 4b — Context Reranker + Conflict Resolver

Before the LLM sees anything, the three streams are merged and cleaned. This is the step most often omitted in naive implementations and the most common source of hallucinations.

- Deduplication: same fact appearing in graph result and semantic result is merged into one entry
- Version conflict resolution: if graph has AMI limit for 2022 and semantic has 2023 for the same MSA/bedroom, the 2023 version wins (more recent). Conflict is logged
- Relevance reranking: a cross-encoder reranker scores all remaining context chunks against the original query and drops low-scoring items
- Context budget enforcement: total context assembled must fit the LLM's context window minus the output budget for the memo section
- Source tagging: every retained fact is tagged [GRAPH], [SEMANTIC], or [COMMUNITY] so the LLM can cite appropriately

Layer 5 — LLM Generation

- Model: Claude claude-sonnet-4-20250514 (current production Sonnet) for memo section generation

- Prompt structure: system prompt defines the underwriting memo section format; user prompt contains the assembled tagged context
- Citation requirement: every factual claim in the output must include a node reference in the form [source: NodeType/node_id] — this is enforced in the system prompt
- Output format: structured per the 13-section memo template; each section is generated independently to keep context focused
- The regulatory hierarchy is referenced explicitly in the system prompt: 'When citing rules, always cite the highest-level authority in the hierarchy'

Layer 6 — Eval Framework

Four evaluation dimensions, each requiring a separate test set and scoring method:

Eval dimension	Method and ground truth source
Graph retrieval accuracy	Curated set of 50+ underwriting questions with known correct graph paths. Measure path-level precision and recall. Ground truth: manually traced paths by domain expert.
Rule accuracy	Factual extraction test: extract numeric values (AMI limits, DSCR thresholds, LTV caps, credit periods) from generated memos and compare against HUD/IRS source tables. Zero tolerance for wrong-year AMI limits.
UW conclusion correctness	Synthetic deal set of 20+ deals with expert-labeled outcomes (approve/condition/decline). Compare system output to expert label. Target: >90% agreement on binary approve/decline.
Citation quality / hallucination rate	Every node reference in output is checked against the graph — does the cited node exist and does it actually support the claim? Hallucination rate = % of claims with no valid supporting node.

- Eval feedback loop: eval failures route back to specific graph refinement tasks — wrong AMI year = data pipeline fix; wrong rule node = ontology fix; hallucinated citation = prompt engineering fix
- Eval runs on every graph update and every model change — not just at initial build

7. Key Architecture & Design Decisions

The following decisions were made explicitly during design and should not be revisited without architectural review.

#	Decision	Choice made	Rationale
1	Graph database	Neo4j (or Kuzu for embedded)	Native graph traversal with Cypher; relationship-first query model matches the LIHTC hierarchy structure
2	Vector store	Pinecone or Weaviate	Managed vector index with metadata filtering; allows filtering by node_type, year, state before kNN
3	Embedding model	text-embedding-3-large or domain fine-tuned variant	Strong on legal/regulatory text; 3072 dimensions; fine-tune on §42 corpus if retrieval accuracy is below threshold
4	Community detection	Leiden algorithm (not Louvain)	Leiden has better resolution stability and handles the heterogeneous LIHTC graph structure more reliably
5	LLM for generation	Claude claude-sonnet-4-20250514	Best citation-following behavior; strong on structured document generation; system prompt supports strict output format enforcement
6	LLM for query planning	Claude Haiku (lighter model)	Query decomposition is a simpler task; using Haiku reduces latency and cost at the query planning step
7	Retrieval strategy	Three-stream: graph + semantic + community	No single stream is sufficient. Graph gives precision; semantic gives recall; community gives global context. All three required for complete UW answers.
8	Simulation anchor	Real HUD LIHTC properties as base	Anchoring simulations to real HUD property records ensures geographic, regulatory, and demographic realism while synthetic financials fill the private data gap
9	Regulatory hierarchy encoding	Directed PRECEDES edges between Regulation nodes	Makes hierarchy machine-traversable, not just a human concept. Prevents child rules from overriding parent rules in graph traversal.
10	Eval cadence	Eval runs on every graph update and model change	LIHTC rules change annually (AMI limits, QCT lists, QAPs). A stale graph is a compliance risk. Continuous eval is non-negotiable.
11	Output citation requirement	Every claim must cite a node ID	Auditable outputs are the core value proposition over naive RAG. Uncited

			claims are treated as hallucinations regardless of correctness.
12	Context window management	Section-by-section generation (not whole memo)	Keeps retrieval focused per section; avoids context dilution; allows independent re-generation of failed sections

8. Existing Players & Prior Art

Research was conducted across SEC filings, patent databases (USPTO/Google Patents), fintech product disclosures, and academic literature. Key findings:

8.1 Closest existing systems

Player / source	What they have and what's missing
ICE / Black Knight (SEC 10-K)	Empower LOS uses AI decisioning; large data network moat across 50M+ loan records. Missing: no graph-structured regulatory hierarchy; no LIHTC-specific ontology; proprietary and closed.
Walker & Dunlop (SEC 10-K + GeoPhy acquisition)	GeoPhy built graph-based property valuation; W&D applies it to affordable housing lending. Missing: valuation focus, not underwriting rationale; no community detection; no regulatory KG.
RiskSpan Edge platform	ML-based LIHTC portfolio risk scoring; closest existing product in this space. Missing: portfolio risk focus (not origination underwriting); no graph-native reasoning; no explainable citation trail.
Blooma AI (acquired by Zions Bank)	CRE underwriting automation with deal scoring that resembles a KG without naming it. Missing: general CRE focus; no LIHTC-specific regulatory hierarchy; no community summaries.
Cherre (PropTech)	Graph-native real estate data linking platform. Missing: data infrastructure play, not an underwriting reasoning system; no §42 regulatory layer.
Novogradac (specialist firm)	Gold-standard LIHTC financial models and rent/income limit databases; used by practitioners. Missing: Excel-based, not an AI or graph system; no RAG or LLM component.

8.2 Patent search guidance

Before building, conduct a patent search at patents.google.com. Key search parameters:

- CPC class G06Q40/02 (mortgage/loan processing) AND 'knowledge graph'
- CPC class G06Q40/02 AND 'tax credit' AND 'underwriting'
- Assignees to watch: CoreLogic, ICE/Black Knight, Fannie Mae, Freddie Mac, MSCI/RCA

As of the architecture design session, no granted patent combining KG-RAG with §42 compliance reasoning was identified. This should be verified before filing any IP.

8.3 The capability gap this system addresses

What no existing system does

No disclosed system combines all three layers:

- (1) A traversable §42 regulatory hierarchy graph with per-node embeddings

(2) Deal-level data anchored to real HUD LIHTC properties with §42-constrained simulation

(3) A three-stream retrieval system that merges graph precision + semantic recall + community benchmarks

(4) LLM-generated underwriting rationale with every claim cited to a specific named graph node

The explainable reasoning chain — 'this deal qualifies for the 30% basis boost because it is in QCT tract 06-037-2653

per HUD Notice 2024-xx, per §42(d)(5)(B)' — is not present in any disclosed product.

9. Recommended Build Sequence

The following sequence is designed to produce a working end-to-end demo as early as possible, then expand scope incrementally. Each phase has a concrete, testable output.

Phase	Name	Deliverable	Key decisions to make
Phase 0	Domain & data setup	Download HUD LIHTC DB, HUD AMI tables, QCT/DDA lists, IRS §42 text. Define LIHTC ontology schema (node types, edge types, rule nodes). Set up graph DB and vector store.	Graph DB choice: Neo4j vs Kuzu. Vector store choice. Embedding model selection.
Phase 1	Graph seed — 5 real properties	Pick 5 real HUD LIHTC properties across 2–3 states. Build complete subgraph for each: Property → Tract (QCT flag) → TaxCreditProgram → AMILimit → Regulation nodes. Populate UW memo sections 1–4 and 11 from public data.	State selection (start with CA + TX for QAP diversity). Confirm ontology completeness.
Phase 2	Regulatory backbone	Ingest full IRC §42 text as Regulation nodes. Ingest 3–5 state QAPs as QAPRule nodes. Build PRECEDES edges for regulatory hierarchy. Validate: can a Cypher query traverse Property → §42(d)(5)(B) correctly?	QAP parsing strategy (PDF → NLP or manual). Hierarchy edge validation.
Phase 3	Single-stream retrieval demo	Wire graph search to LLM. Given a real property, generate sections 1–4 and 11 with cited node references. Demonstrate: 'This property qualifies for QCT boost per [Tract node] → [§42(d)(5)(B) Regulation node]'	Prompt template for section generation. Citation format standard.
Phase 4	Semantic stream + embeddings	Generate per-node embeddings for all graph nodes. Stand up vector store index. Implement kNN retrieval for AMILimit and Regulation nodes. Demonstrate: semantic query finds correct AMI limit node for given MSA/year/BR.	Embedding model fine-tuning decision. Vector store metadata schema.
Phase 5	Simulation layer	Build §42-constrained synthetic data generator for LoanFile, CapitalStack, ProForma nodes. Anchor to 5 real HUD properties. Validate: DSCR > 1.20, LTV within GSE cap, dev fee within QAP cap.	Simulation parameter ranges. Validation rules for generated data.
Phase 6	Community detection	Run Leiden clustering on full graph. Identify 3–5 natural communities. Generate pre-computed community summaries. Implement community retrieval stream.	Clustering resolution parameter. Community summary prompt template.
Phase 7	Three-stream + reranker	Implement query planner (decompose → route). Implement all three retrieval streams. Build context reranker and	Reranker model choice. Context budget allocation across 3 streams.

		conflict resolver. Demonstrate full pipeline on 10 UW questions.	
Phase 8	Eval framework	Build eval test sets for all 4 dimensions. Run baseline evals on Phase 7 system. Identify top failure modes. Establish eval CI pipeline.	<i>Ground truth labeling process. Eval pass/fail thresholds.</i>
Phase 9	Scale + refine	Expand to 50+ properties across 10+ states. Add annual AMI limit refresh pipeline. Refine graph and prompts based on eval results.	<i>Data refresh cadence. Graph versioning strategy.</i>

10. Team Roles & Study Guidance

Different team members need to internalize different parts of this document. The following guidance maps roles to study priorities.

Role	Priority sections to study deeply
Graph / data engineer	Section 6 (Layer 0–2 detail): ontology, embedding strategy, Leiden community detection. Section 3: data sources and download instructions. Phase 0–2 of Section 9.
ML / NLP engineer	Section 6 (Layer 3–4b): query decomposition, three-stream retrieval, reranker. Section 6 Layer 0 embedding model decisions. Sections 7 decisions #3, 4, 7.
LLM / prompt engineer	Section 6 (Layer 5): generation prompt structure, citation requirements, context budget. Section 6 (Layer 6): eval framework, hallucination rate metric. Section 7 decisions #5, 11, 12.
Domain expert (LIHTC)	Section 2: regulatory hierarchy — validate completeness and correctness. Section 4: UW memo sections — validate section descriptions and fill strategies. Section 6 Layer 6 eval: define ground truth for UW conclusion correctness.
Product / program manager	Section 1: scope and objectives. Section 8: competitive context. Section 9: full build sequence and decisions to make at each phase. Section 7: architecture decisions log.
Architect / tech lead	All sections. Specifically Section 5 (architecture overview), Section 7 (decisions), and the gaps analysis in Section 5 callout box. Responsible for resolving open decisions at each phase gate.

10.1 Open questions requiring team input before Phase 0

1. Graph DB choice: Neo4j (managed, mature ecosystem, Cypher) vs Kuzu (embedded, faster for smaller graphs, open source). Decision depends on deployment environment.
2. Embedding model: use OpenAI text-embedding-3-large out of the box, or fine-tune a domain model on \$42 text corpus? Fine-tuning adds 2–3 weeks but may significantly improve regulatory retrieval accuracy.
3. State selection for Phase 1: recommend California (largest LIHTC market, complex QAP) and Texas (different QAP structure, different market conditions). Team to confirm.
4. Simulation validation process: who is the domain expert who will validate that synthetic loan files are structurally realistic? This person is critical for Phase 5.
5. Eval ground truth: who will manually label the 20+ synthetic deal test set for UW conclusion correctness (approve/condition/decline)? This requires LIHTC underwriting expertise.
6. Data refresh cadence: HUD AMI limits and QCT lists update annually (typically March). Graph update pipeline must be designed before Phase 9 but the cadence decision is needed early.

Appendix A — Glossary

Term	Definition
AMI	Area Median Income. Published annually by HUD by MSA/county. Determines maximum tenant income and maximum restricted rent for LIHTC units.
Community (graph)	A cluster of densely connected nodes identified by Leiden/Louvain clustering. Enables global 'peer cohort' context retrieval.
DDA	Difficult Development Area. HUD-designated high-construction-cost areas. Grants the same 30% eligible basis boost as QCT.
DSCR	Debt Service Coverage Ratio. $\text{Net Operating Income} / \text{Annual Debt Service}$. GSE minimum typically 1.20.
FMR	Fair Market Rent. HUD-published annual market rent estimates by MSA and bedroom size. Used as market rent baseline.
Form 8609	IRS Low-Income Housing Credit Allocation and Certification. Issued by state HFA; formally allocates the tax credit to a property.
Form 8823	IRS Low-Income Housing Credit — Agencies Report of Noncompliance. Filed by state HFA when a compliance issue is detected.
Form 8586	Low-Income Housing Credit. Filed by the taxpayer/investor annually to claim the credit.
GSE	Government Sponsored Enterprise. Fannie Mae and Freddie Mac. Their program guides set underwriting parameters for loans they purchase or guarantee.
KG-RAG	Knowledge Graph Retrieval Augmented Generation. A RAG architecture where retrieval follows graph relationships rather than pure vector similarity.
Leiden algorithm	Community detection algorithm preferred over Louvain for resolution stability. Groups densely connected graph nodes into clusters.
LIHTC	Low Income Housing Tax Credit. The primary US federal program for financing affordable rental housing, governed by IRC §42.
LTV	Loan-to-Value ratio. $\text{Loan amount} / \text{property value}$. FHA 223(f) caps at approximately 87% for affordable housing.
LURA	Land Use Restriction Agreement. Deal-level recorded document binding the property to income and rent restrictions, typically for 30+ years.
Multi-hop traversal	Graph query pattern that follows multiple relationship edges in sequence. E.g., $\text{Property} \rightarrow \text{CensusTract} \rightarrow \text{QCT flag} \rightarrow \text{Regulation node}$.
NOI	Net Operating Income. $\text{Gross rental income} - \text{vacancy} - \text{operating expenses}$. The numerator of DSCR.
Per-node embedding	A vector representation of an individual graph node, enabling semantic similarity search over graph entities.

PLR	Private Letter Ruling. IRS guidance issued to a specific taxpayer. Cited in graph as a Regulation node.
QAP	Qualified Allocation Plan. Annual document published by each state Housing Finance Agency governing how LIHTC credits are allocated.
QCT	Qualified Census Tract. HUD-designated high-poverty census tracts. Properties located in QCTs receive a 30% boost to eligible basis under §42(d)(5)(B).
RAG	Retrieval Augmented Generation. LLM architecture where relevant context is retrieved from an external knowledge source before generation.
Reranker	A cross-encoder model that scores retrieved context chunks against the original query to select the most relevant items before LLM generation.
TAM	Technical Advice Memorandum. IRS guidance to an IRS employee on a specific case. Cited in graph as a Regulation node.

Appendix B — Public Data Source Reference

Source	URL / access method
HUD LIHTC Database	https://www.huduser.gov/portal/datasets/lihtc.html — annual CSV download; 50,000+ properties since 1987
HUD Income Limits (AMI)	https://www.huduser.gov/portal/datasets/il.html — annual Excel/CSV by state, county, household size
HUD QCT and DDA Lists	https://www.huduser.gov/portal/datasets/qct.html — annual Excel files; match to census tract FIPS codes
HUD Fair Market Rents	https://www.huduser.gov/portal/datasets/fmr.html — annual by MSA and non-metro area, bedroom count
IRS §42 and related	https://www.irs.gov/credits-deductions/businesses/low-income-housing-credit — links to Rev. Procs., Notices, Form instructions
IRS full tax code	https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title26-section42 — IRC §42 full text
State HFA QAPs	Each state HFA website — search '[state] housing finance agency qualified allocation plan [year]'. All 50 states publish publicly.
FFIEC HMDA data	https://ffiec.cfbp.gov/data-browser/ — loan-level origination data by census tract, lender, loan type
Fannie Mae DUS Guide	https://multifamily.fanniemae.com/node/2211 — Selling and Servicing Guide for DUS lenders
Freddie Mac TAH	https://mf.freddiemac.com/investors/tax-exempt-bonds/tax-credit-affordable.html — TAH program documentation
SEC EDGAR full-text search	https://efits.sec.gov/LATEST/search-index?q=%22LIHTC%22+%22machine+learning%22&dateRange=custom&startdt=2020-01-01 — search 10-K filings for AI/ML LIHTC disclosures
Google Patents	https://patents.google.com/?q=knowledge+graph+mortgage+underwriting&before=priority:20260101 — prior art search
Urban Institute Housing Finance	https://www.urban.org/policy-centers/housing-finance-policy-center — research papers on affordable housing data and AI

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This document is the primary onboarding reference. All architectural decisions recorded herein supersede prior diagrams.